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Stability and Habitability Overlap in Hierarchical Triple Systems: Influence of Companion Orbital Parameters

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ABSTRACT

This Research Training report presents the second stage of a study on planetary stability and habitability in hierarchical triple star systems. Carried out at the Institut de Planétologie et d'Astrophysique de Grenoble (IPAG), this project transitions from the theoretical framework established in the first report to a numerical exploration. The study aims to identify the overlap between dynamical stability regions and radiative habitable zones. Using the `REBOUND` N-body integrator, we investigate the influence of the outer companion's mass (m_C) and eccentricity (e_C) on the survival of circumbinary planets. Our results show that the companion's gravity is a primary constraint on planetary survival, with increased mass and eccentricity significantly reducing the size of the stable zone.

Key words. methods: numerical – methods: N-body simulations - celestial mechanics - planetary habitability and stability - multiple star system

1. Methods and physical models

This section describes the numerical framework used to investigate how the eccentricity and mass of an outer stellar companion affect the orbital stability and radiative habitability of a circumbinary planet in a hierarchical triple star system. We use N-body simulations performed with the Python package `REBOUND` (Rein & Liu 2012) to follow the long-term orbital evolution of the system, and we compute the time-dependent radiative environment experienced by the planet in order to identify the regions where dynamical stability and habitability overlap.

1.1 Dynamical model

According to the nomenclature convention for hierarchical multiple systems (Hessman et al. 2010), the two stars of the inner binary are denoted A and B, and the binary itself is referred to as AB. The outer companion star is denoted $(AB)C$, which will be simplified as C. Finally, the circumbinary planet is denoted $(AB)b$, which will be simplified as p.

In this work, we restrict ourselves to coplanar hierarchical triple star systems with only prograde orbits. Such systems are composed of an inner binary around which the planet and an outer companion star orbit the barycenter of the binary, inducing gravitational perturbations on the planet. This configuration implies that the companion's orbital period is much longer than the binary's orbital period (Verrier & Evans 2007). This separation favors the long-term stability of the system compared to a non-hierarchical triple system, where the three-body motion is intrinsically chaotic.

Accordingly, the hierarchical and coplanar assumptions can be written as shown in Equation 1.

$$P_{AB} \ll P_C\,; \qquad i_{AB} = i_C = i_p = 0 \tag{1}$$

where P_{AB} is the orbital period of the inner binary, P_C is the orbital period of the outer companion around the barycenter of the inner binary, i_{AB} is the inclination of the inner binary orbit, i_C is the inclination of the companion, and i_p is the inclination of the planet. See Appendix A for a reminder of the orbital elements used throughout this work.

Our objective is to isolate the influence of the companion mass m_C and eccentricity e_C on the dynamical stability and radiative environment of the circumbinary planet. To this end, all other system parameters are kept fixed:

$$m_A = 2\,M_\odot\,; \quad m_B = 1\,M_\odot\,; \quad e_{AB} = 0\,; \quad a_{AB} = 0.1\,\mathrm{AU}\,; \quad i_{AB} = 0$$
$$a_C = 15\,\mathrm{AU}\,; \quad i_C = 0$$
$$m_p = 1 M_\oplus\,; \quad e_p = 0\,; \quad i_p = 0$$

From this reference configuration, we vary the companion mass m_C from 0.1 to $2.3\,M_\odot$ and its eccentricity e_C from 0 to 0.6. We then introduce the following dimensionless parameter to take into account the role of the binary mass on stability:

$$\mu_C = \frac{m_C}{m_{AB}} = \frac{m_C}{m_A + m_B}$$

Accordingly, μ_C varies from 0.03 to 0.77, following the adopted range of m_C.

Finally, in order to avoid bias from any particular initial condition, we perform N random realizations of the initial phase of the system. A given planetary semi-major axis a_p is considered stable if at least a fraction $q = 0.9$ of these initial phase realizations lead to a stable planetary orbit over $n = 10^4$ periods. This condition can be written as follow:

$$p(a_p, (\mu_C, e_C)) = \frac{N_{\mathrm{stable}}}{N} > q = 0.9 \tag{2}$$

We then sample a range of k planetary semi-major axes a_p within the circumbinary region, defines as the range of a_p between a_{AB} and a_C, in order to determine the inner and outer boundaries of the stability zone, or equivalently to construct the stability bar graph. The parameter k therefore sets the spatial resolution of the stability zone.

1.2 Numerical integrator

The orbital evolution of the system is computed with the REBOUND N-body Python package (Rein & Liu 2012), using the symplectic integrator WHFast (Rein & Tamayo 2015). REBOUND is an open-source code for gravitational dynamics, available in both C and Python for ease of use. This choice is motivated by the fact that symplectic integrators are particularly well suited for long-term integrations of planetary systems, as they provide good conservation of the total energy over long timescales. In addition, other integrators, such as the 15th-order integrator IAS15, can achieve errors below machine precision over at least 10^9 orbital periods, but our computational limitations led us to choose WHFast for its greater computational efficiency.

The integration time step is set to

$$\Delta t = \frac{P_{AB}}{20} \tag{3}$$

where P_{AB} is the orbital period of the inner binary, that is, the shortest orbital period of the system. This choice ensures that the fastest orbital motion is sufficiently resolved throughout the integration. Since planetary stability is intrinsically a time-dependent property, integrating the system over $\sim 10^4$ orbital periods of the relevant body provides a reasonable numerical test to distinguish clearly unstable configurations from long-lived regular ones (Holman & Wiegert 1999). Ideally, each simulation is therefore performed over a total duration of

$$t_{\max} = 10^4 P_p \tag{4}$$

where P_p is the orbital period of the planet. However, due to computational constraints, we adapted this integration time in section 2. In practice, this orbital timescale is estimated using the Keplerian approximation:

$$P_p = 2\pi \sqrt{\frac{a_p^3}{G(m_A + m_B)}} \tag{5}$$

where a_p is the planetary semi-major axis, and $m_A + m_B$ is the total mass of the inner binary. This approximation is adopted because, for perturbed circumbinary trajectories, the orbital period returned by REBOUND may become poorly defined.

A configuration is classified as unstable if the planetary eccentricity reaches or exceeds one:

$$e_p \geq 1 \tag{6}$$

which corresponds to a non-bound, hyperbolic trajectory and is interpreted here as an ejection from the system. Alternatively, a configuration is also considered unstable if the planetary semi-major axis exceeds a prescribed critical distance from the inner binary barycenter, taken as a multiple r of the companion semi-major axis:

$$a_p > r\, a_C \tag{7}$$

where a_C is the semi-major axis of the outer companion and r is a threshold factor. This second condition is introduced to identify strongly perturbed configurations in which the planet is displaced far from the circumbinary region and can no longer be regarded as dynamically associated with the inner binary.

Within this framework, a planet is considered stable if neither of these conditions is met over the full integration time $t_{\max}$. Equivalently, the orbit is considered as stable if it survives the complete numerical integration over 10^4 planetary periods without undergoing escape according to the adopted criterion.

We do not distinguish here between the different physical channels leading to instability, such as ejection or collision, because the bodies are treated as dimensionless particles. In particular, collision events are not treated separately, as close encounters with a star effectively remove the planet from the circumbinary configuration through the strong gravitational acceleration they induce and are therefore counted within the same category as ejections. The purpose of this work is to identify the domain of long-term orbital survival rather than to establish a detailed classification of unstable outcomes.

1.3 Radiative model

To simplify our model as much as possible, we adopt the definition of the Habitable Zone (HZ) introduced by Kasting et al. (1993). This is defined as the region where temperature conditions allow the presence of liquid water on the planetary surface. Consequently, a planet is considered to be within the HZ if its equilibrium temperature remains between 0°C and 100°C (273.15 K to 373.15 K).

However, in multiple star systems, temperature alone is not sufficient. As discussed in our previous work, planetary stability is a prerequisite for long-term habitability. Since we integrate the system over 10^4 planetary orbital periods to ensure dynamical stability, the planet must also remain within the HZ boundaries throughout the entire duration of the simulation.

For this reason, we will use the concept of the Permanently Habitable Zone (PHZ) introduced by Eggl et al. (2012). Unlike the standard HZ, the PHZ specifically accounts for the orbital variations of the planet: it is the region where the planet's entire trajectory stays within the HZ limits, despite the gravitational perturbations and flux variations caused by the binary or triple stellar components.

We developed a Python program to calculate the equilibrium temperature at any point in the orbital plane. To do so, we made the following assumptions:

- The planet is in radiative equilibrium, meaning the energy absorbed is equal to the energy emitted;
- The planet is assumed to emit radiation as a perfect blackbody;
- The absorbed energy is redistributed over the entire planetary surface before being re-emitted. This assumes that the rotation period is much shorter than the orbital period, leading to a uniform temperature and a global emitting area ($4\pi R_p^2$);
- Radiative properties, specifically the albedo, are independent of the wavelength. In our model, we assume an Earth-like albedo of $\alpha = 0.3$;
- The planet is assumed to have an Earth-like atmosphere. To account for the greenhouse effect, an empirical correction factor of $f_{gh} = 1.13$ (see Appendix B) is applied to the equilibrium temperature.
- Mutual shadowing and eclipses between the stars and the planet are neglected;
- The planetary cross-section intercepting the stellar flux is considered a disk (πR_p^2);
- The stars' luminosity remains constant throughout the integration time. This is justified as the simulation duration (10^4 orbital periods) is negligible compared to the timescales of stellar evolution.

In order to find the thermal radiation density received by the planet, we need the stars' outgoing flux. We can obtain this using

the mass-luminosity relation below (Duric 2004):

$$L_i = \begin{cases} 0.23\, L_\odot \left(\frac{m_i}{M_\odot}\right)^{2.3} & \text{if } m_i < 0.43\, M_\odot \\ L_\odot \left(\frac{m_i}{M_\odot}\right)^{4} & \text{if } m_i \geq 0.43\, M_\odot \end{cases} \quad (8)$$

where L_i and M_i are the luminosity and the mass of the star i, and $L_\odot$ and $M_\odot$ are the luminosity and the mass of the Sun.

We can determine the planet's equilibrium temperature T_{eq} by balancing the emitted and received fluxes, according to Stefan-Boltzmann's law. The energy balance is expressed as follows:

$$4\pi R_p^2 \times \sigma T_{eq}^4 = \pi R_p^2 \times \frac{1}{4\pi} \left(\sum_i \frac{L_i}{d_i^2} \right) (1 - \alpha) \quad (9)$$

where d_i is the distance between the planet and the star i, and σ is the Stefan-Boltzmann constant. By isolating the temperature and applying the correction factor f_{gh}, we obtain:

$$T_{eq} = f_{gh} \left[\frac{1 - \alpha}{16\pi\sigma} \left(\sum_i \frac{L_i}{d_i^2} \right) \right]^{1/4} \quad (10)$$

In practice, we implemented these equations in a Python script to generate temperature maps.

Based on the stellar masses, the program first determine the luminosity of the stars, thanks to Equation 8. Next, we create a two-dimensional grid on the orbital plane. At each step of the simulation, the program computes the distance d_i from each cell of the grid to the three moving stars. Finally, we sum the fluxes from each source, and we convert it to the equilibrium temperature T_{eq} thanks to Equation 10.

Now that we have calculated the temperature map, we can define the HZ. Because of the motion of the stars, the HZ is not a static region. For the sake of clarity, it is more convenient to use PHZ, which is not time dependent. For that reason, our script integrates the temperature maps over the entire REBOUND simulation (10^4 orbital planetary periods). By the end of the simulation, the program can identify the cells that are part of the PHZ by filtering the points where, for each time step of the simulation, the HZ condition is satisfied.

2. Results

2.1 Figure presentation

Due to the high computational cost of the simulations, we were unable to test planetary stability over the ideal $n = 10^4$ orbital periods (P_p). We acknowledge that this limitation may lead to a slight overestimation of the stability zone; however, it still provides a reliable overview of the system's dynamics. Therefore, we performed our initial simulation using the highest precision parameters computationally feasible for our setup: stability is evaluated over $n = 10^3$ planetary periods, using a spatial resolution of $k = 30$ (dividing the space between the inner binary and the outer companion into 30 discrete distances), and testing $N = 30$ randomized initial phases for each distance.

Figure 1 shows the stability fraction as a function of the planetary semi-major axis for the configuration $e_C = 0$ and $m_C = 1.5\, M_\odot$. Each bar provides the fraction of stable realizations obtained for a given value of a_p, based on different initial orbital phases. The vertical dashed line marks the semi-major axis of the outer stellar companion.

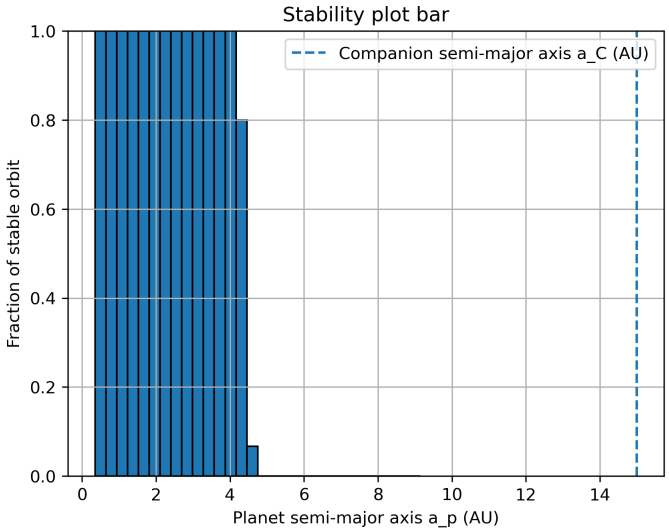


Fig. 1: Stability bar plot for an Earth-like planet with $e_C = 0$ and $m_C = 1.5\, M_\odot$

The results reveal a broad, stable circumbinary region extending from 0.5 AU to 4.3 AU.

Figure 2 presents a two-dimensional temperature map of the system at one representative time step, together with the Stability Zone (SZ) and the Permanently Habitable Zone (PHZ). The color map shows the spatial distribution of the equilibrium temperature produced by the three stars, while the blue annular region marks the PHZ. The green circles indicate the inner and outer boundaries of the dynamically stable zone. A planet is placed at $a_p = 3.3$ AU with a blue line showing its orbit.

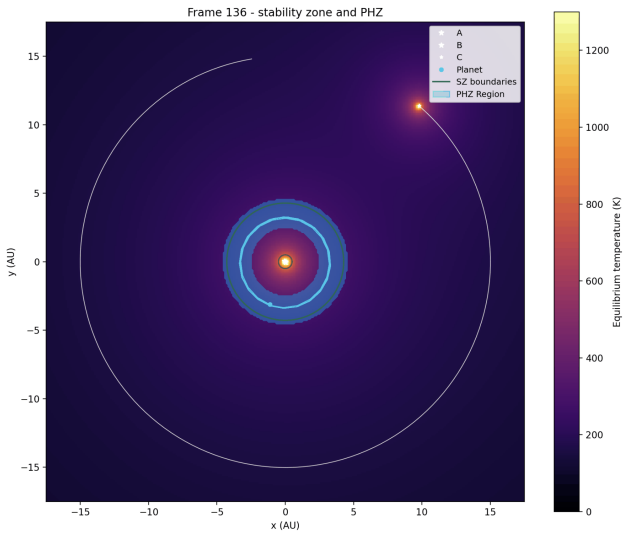


Fig. 2: Temperature map for an Earth-like planet orbiting at $a_p = 3.3$ AU and showing the SZ and PHZ

This figure provides a direct visualization of the overlap between dynamical and radiative constraints.

Figure 3 shows the time evolution of the planetary equilibrium temperature for the case $e_C = 0$ and $m_C = 1.5\, M_\odot$, with the planet placed at $a_p = 3.30$ AU. The black curve represents the instantaneous surface equilibrium temperature of the planet,

while the blue shaded region indicates the liquid-water temperature interval adopted here as the habitable range.

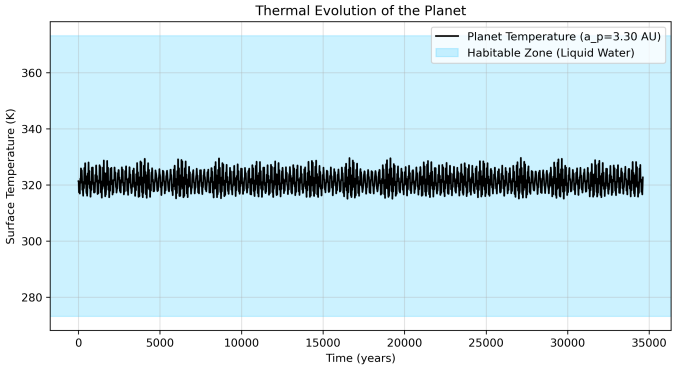


Fig. 3: Surface temperature over $10^4 P_p$ of an Earth-like planet at $a_p = 3.3$ AU with $e_C = 0$ and $m_C = 1.5\ M_\odot$

2.2 Companion study

We summarize in Table 1 and Table 2 the data obtained from the analysis of the companion eccentricity and mass. We used a lower integration time, $n = 10^3\ P_p$, to reduce computing time. This shorter integration time may lead to an overestimation of the Stability Zone. For each explored configuration, we report the inner and outer boundaries of the Stability Zone (SZ), together with the number of grid cells belonging to the Permanently Habitable Zone ($N_{\rm PHZ}$). The SZ boundaries provide a direct measure of the radial extent of the dynamically stable circumbinary region, while $N_{\rm PHZ}$ serves as a simple numerical proxy for the size of the permanently habitable domain in the temperature maps. Table 1 presents the dependence of these quantities on the companion mass m_C for a fixed eccentricity $e_C = 0$, whereas Table 2 shows their variation with companion eccentricity e_C for a fixed mass $m_C = 0.1\ M_\odot$. These tabulated values make it possible to compare more clearly how the companion's mass and orbital eccentricity affect both the dynamical stability and the radiative habitability of the system.

Table 1: Stability Zone boundaries (SZ) and number of points in the Permanently Habitable Zone ($N_{\rm PHZ}$) as a function of companion mass m_C, for a fixed eccentricity $e_C = 0$.

$m_C\ (M_\odot)$	μ_C	SZ (AU)	$N_{\rm PHZ}$
0.1	0.03	[0.8, 8.1]	3288
0.3	0.10	[0.8, 6.9]	3288
0.5	0.17	[0.8, 6.5]	3288
0.8	0.27	[0.8, 5.7]	3342
1.0	0.33	[0.8, 5.7]	3342
1.3	0.43	[0.8, 4.7]	3342
1.6	0.53	[0.8, 4.2]	3378
2.0	0.67	[0.8, 4.2]	3477
2.3	0.77	[0.8, 4.2]	3620

Due to our limited computational resources, the adopted numerical resolution affects our results and likely explains the step-like behavior observed in N_{PHZ} and the edges of the stability zone.

Overall, our simulations show that the hierarchical triple systems studied here can preserve a significant overlap between dynamical stability and radiative habitability. In the configurations explored here, the Stability Zone remains sufficiently extended to host circumbinary planetary orbits over long timescales, while part of this region is also compatible with surface temperatures allowing liquid water. The temperature evolution of a representative planet further shows that the thermal environment can remain relatively regular, with bounded and periodic variations and no extreme long-term drift. Overall, these results suggest that such systems may still provide favorable conditions for long-term planetary survival and surface habitability, although the extent of this overlap strongly depends on the orbital and physical properties of the stellar companion.

Table 2: Stability Zone boundaries (SZ) and number of points in the Permanently Habitable Zone ($N_{\rm PHZ}$) as a function of companion eccentricity e_C, for a fixed companion mass $m_C = 0.1\ M_\odot$.

e_C	SZ (AU)	$N_{\rm PHZ}$
0.00	[0.8, 8.1]	3288
0.20	[1.1, 6.3]	3288
0.40	[1.6, 4.9]	3288
0.60	[2.0, 4.0]	3288

Our results indicate that the companion mass and eccentricity do not affect the system in the same way. Increasing the companion mass m_C tends to reduce the radial extent of the stability zone, while slightly enlarging the Permanently Habitable Zone. In this case, the temperature variations induced by a massive companion remain periodic and of relatively small amplitude (see Figure 3), so that the thermal environment stays fairly regular and compatible with permanent habitability. By contrast, increasing the companion eccentricity strongly reduces the size of the stability zone, while leaving the PHZ nearly unchanged. However, eccentric companions tend to induce much larger temperature variations on the planet (see Figure 4), leading to a more variable and less stable thermal environment. Although the radiative habitable region may still exist geometrically, these strong thermal variations make such configurations less favorable to the emergence and long-term maintenance of life. We can also see that increasing the companion eccentricity tends to compress both the inner and outer boundaries of the stability zone, whereas increasing the companion mass is affected mainly through a reduction of its outer boundary.

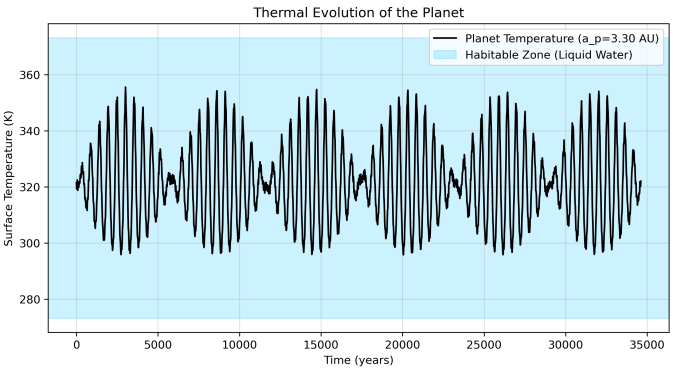


Fig. 4: Surface temperature over $10^4 P_p$ of an Earth-like planet at $a_p = 3.3$ AU with $e_C = 0.4$ and $m_C = 0.1\ M_\odot$

3. Conclusion

Although our results overestimate the stability zone (because of the short integration time), depend heavily on spatial resolution,

and lack robustness, we have been able to identify the main influences on the stability and habitability of an exoplanet in such a system.

Our numerical results show that the outer companion characteristics really influence the planet's stability and habitability. More precisely, if we increase the companion's mass, it mostly reduces the outer limit of the Stability Zone (SZ), even if the temperature remains quite regular. On the contrary, a higher companion eccentricity strongly reduces both the inner and outer limits of the SZ. It also creates very large temperature changes for the planet. In the end, both parameters greatly reduce the space where dynamical stability and the Permanently Habitable Zone (PHZ) can overlap.

To improve this project, several technical adjustments could be made. Increasing the spatial resolution (k) and the number of random initial phases (N) would lead to more precise and robust stability boundaries. Furthermore, integrating the orbits over 10^4 planetary periods instead of 10^3 would provide a more accurate assessment of long-term stability. To make these high-resolution simulations feasible, future work should also focus on code optimization, such as implementing parallel processing, to significantly reduce computational costs and execution times.

Future work could expand this project to the study of more complex scenarios. This includes exploring non-coplanar systems to account for the impact of orbital inclination and the Kozai-Lidov mechanism. Additionally, incorporating more sophisticated atmospheric models would allow for a more detailed estimation of surface temperatures and habitability limits beyond the simple greenhouse factor used in this study. We could also work with different planetary profiles, such as Venus, which has a different albedo than Earth. This would allow us to see the impact of the characteristics of a planet on the Habitable Zone.

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Appendix A: Orbital parameters

The orbit of a body around a central object is described by a set of orbital parameters that define the shape of the orbit, its orientation in space, and the motion of the body along it (see Figure A.1):

- Inclination i: measures the tilt of the orbital plane relative to a reference plane. If $i = 0°$, the orbit is coplanar and prograde, whereas if $i = 180°$, it is coplanar and retrograde. More generally, an orbit can be characterized as prograde or retrograde relative to the other bodies in the system, depending on whether its inclination lies between 0° and 90°, or between 90° and 180°.
- Longitude of the ascending node Ω : specifies the direction of the ascending node, that is, the point where the orbital plane crosses the reference plane.
- Argument of the periapsis ω: defines the orientation of the ellipse within the orbital plane.
- True anomaly ν or f: gives the instantaneous position of the body along its orbit, measured from the periapsis.
- Semi-major axis a: defines the size of the orbit, it corresponds to half of the longest diameter of the orbital ellipse.
- Eccentricity e: defines the shape of the orbit by quantifying how much it deviates from a circle. For $e = 0$, the orbit is circular, while for $0 < e < 1$, it is elliptical, with larger values of e corresponding to more elongated ellipses. If $e > 1$, the body is no longer gravitationally bound to the central object, and its trajectory is hyperbolic.
- Orbital period P: defines the time required for a gravitationally bound body to complete a full orbit around its central object.

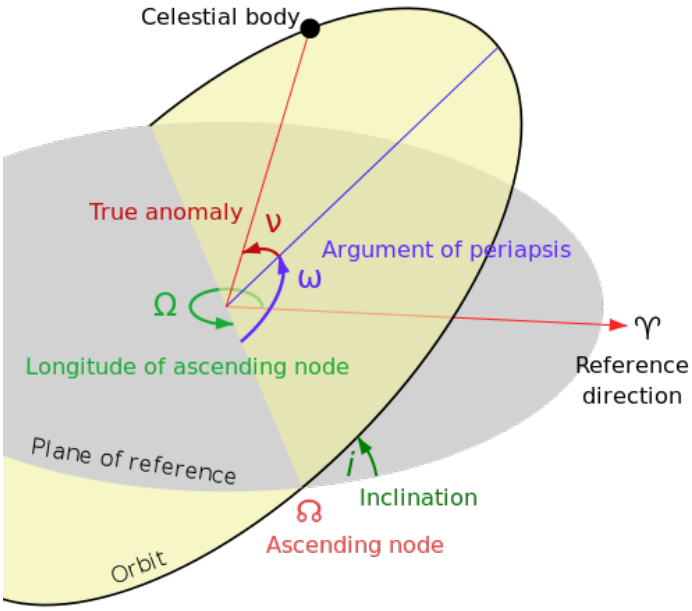


Fig. A.1: Geometrical representation of the orbital elements according to the reference plane in grey. Source: Rein (2026).

Appendix B: Calculation of the greenhouse factor f_{gh}

To estimate a realistic surface temperature, it is necessary to account for the warming effect of the planetary atmosphere. We first define the effective equilibrium temperature of the Earth ($T_{\text{eff},\oplus}$), which represents the theoretical temperature of the planet without an atmosphere. According to Equation 10 (with $f_{gh} = 1$), we have:

$$T_{\text{eff},\oplus} = \left[\frac{1-\alpha}{16\pi\sigma}\left(\frac{L_\odot}{d_{E-S}^2}\right)\right]^{1/4} \approx 255 \text{ K} \tag{B.1}$$

where d_{E-S} is the distance between the Earth and the Sun.

However, the actual mean surface temperature of the Earth is measured to be $T_{\text{surf},\oplus} = 288$ K (Kasting et al. 1993). This difference of 33 K is attributed to the natural greenhouse effect of our atmosphere. By comparing the observed value with the theoretical one, we determine the following empirical correction factor:

$$f_{gh} = \frac{T_{\text{surf},\oplus}}{T_{\text{eff},\oplus}} \approx 1.13 \tag{B.2}$$

In our numerical model, this factor is applied to the equilibrium temperature. It provides an approximation of the surface temperature for Earth-like planets.

Appendix C: Code and Supplementary Material

The Python script written for this study is available on GitHub, click here. Along with the source code, you will also find:

- Our first Research Training report, which provides the theoretical framework and literature review for this project.
- Animation of the temperature maps showing the Stability Zone and the Permanently Habitable Zone.
- Other example figure